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EXAMINER
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NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/743,967	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2457	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 08/24/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 06/14/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims*** 1-20 are pending.

2. ***Claims*** 1-20 ***are rejected.***

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

3. ***Claim***(s) 1 and 11 is/are rejected under 35 U.S.C. 103 as being unpatentable over Oddo et al (Patent. No.: US 10,296,862 A1) in view of Koda et al (Pub. No.: US 2018/0113460) and Levinson et al (Pub. No.: US 2017/0123428).

As per claim 1, Oddo discloses a system, comprising:

- ***a vehicle*** (Oddo, col 1 lines: 46-48, wherein "This disclosure relates to a system and method to dynamically evaluate mission plans for both *manned and unmanned vehicles* over selected current or future time periods")***~~having a network comprising a plurality of end points~~;***

- ***a controller*** (Oddo, col 10 lines: 47-49, wherein "Such method can be executed by various components configured in an integrated circuit, processor, or a controller, for example")***, comprising:***

- ***an automation management circuit structured to provide an automated action plan*** (Oddo, col 5 lines: 58-65, wherein "The system 100 employs at least one external mission planner 102 that processes one or more mission plan inputs 105 and generates a mission

definition file (MDF) 120 that includes mission data that describes one or more tasks and routes for one or more vehicles. As shown, the mission plan inputs 105 can also include task priorities and constraints") ***that includes data description of data to be collected identifying required sensor data, the data description including at least one of: a network address, an end point identifier, a resolution, or a sampling rate associated with the required sensor data*** (Oddo, col 6 lines: 5-20, wherein "Task type scoring criteria 134 are defined generically for each type of task planned and may be further constrained by optional inputs accompanying the tasking 105. For example, the general task type scoring criteria 134 for an image collection task may include items such as image target location and a required image quality whereas the general task type scoring criteria for an area search task may include a time of arrival, search duration, and geographical area within which to search"; col 7 lines: 37-41, wherein "As such, a planned imaging task that positions the vehicle carrying the imaging sensor to within a certain range of the target should produce an image with a predictable resolution that can be compared to the resolution criteria specified for that task")***;***

- ***an automation execution circuit structured to provide an automation command in response to the automated action plan*** (Oddo, col 3 lines: 64-67, wherein "When a final plan has been selected, the operator may save it to the plan archive for future execution or issue it directly to a vehicle for immediate execution"; wherein the immediate execution of the final plan inherently teaches providing at least one command to execute the plan).

Oddo does not explicitly disclose that

- the vehicle ***having a network comprising a plurality of end points;***
- ***to evaluate availability of the required sensor data identified by the data description from an end point of the network before executing an automated vehicle response associated with the automation command; and***

- ***an automation continuity circuit structured to determine whether an automation interruption event has occurred, and to provide an automation continuity command in response to the automation interruption event, wherein the automation continuity command includes instructions to delay, modify, or reschedule operations when the required sensor data identified by the data description is unavailable from the end point of the network***. However, Koda discloses

- the vehicle ***having a network comprising a plurality of end points*** (Koda, Fig 2, paragraph 0038-0039, wherein "FIG. 2 is a block diagram indicating a functional configuration of a driving assistance device 1. As illustrated in FIG. 2, the driving assistance device 1 mainly includes a communication unit 11, a storage unit 12, a sensor unit 13, an input unit 14, a control unit 15 and an output unit 16. Under the control of the control unit 15, the communication unit 11 acquires the map data D1 from the server device 2 to thereby register the acquired map data D1 on the partial map DB 20. In this case, for example, at the time of approaching or making a prediction of entering such an area that the corresponding map data has not been registered yet on the partial map DB 20, the communication unit 11 sends information indicating the area to the server device 2 to thereby acquire the map data D1 corresponding to the area. Additionally, under the control of the control unit 15, the communication unit 11 acquires weather information from a server device that delivers weather information")***;***

- ***to evaluate availability of the required sensor data identified by the data description from an end point of the network before executing an automated vehicle response associated with the automation command*** (Koda, Fig 6 step S203-S204 paragraph 0073, wherein "the autonomous driving function determination unit 55 recognizes a set of the autonomous driving function Fc and the automation level Lc which can be performed in the present traveling section and the present traveling lane (step S203). Then, the autonomous driving control unit 57

performs autonomous driving by performing the feasible autonomous driving function(s) Fc determined at step S203 at the feasible automation level Lv (step S204)").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo with Koda to identify sensors and sensors' capabilities required to perform a plan/function because this would improve the reliability and safety of the system by informing the driver and/or the person on interest about the ability to execute the plan ahead of the time of execution.

Oddo and Koda do not explicitly disclose

- ***an automation continuity circuit structured to determine whether an automation interruption event has occurred, and to provide an automation continuity command in response to the automation interruption event, wherein the automation continuity command includes instructions to delay, modify, or reschedule operations when the required sensor data identified by the data description is unavailable from the end point of the network***. However, Levinson discloses

- ***an automation continuity circuit structured to determine whether an automation interruption event has occurred*** (Levinson, paragraph 0152, wherein "In an embodiment, a sensor anomaly may be detected upon the AV system determining that sensor data gathered from the sensor includes a range of laser return intensities that are a result of the reflectivity properties various phenomena, such as too much sunlight, directly or indirectly bouncing off glass, water, or other shiny surfaces, and other lights, such as headlights or external lights in the infrared range interfering with the laser returns. Various conditions, such as weather condition, refraction due to differences in air density due to heat waves off a hot road surface, and glass surfaces with water (e.g., windshields, foggy and/or wet windshields) may cause interferences with LIDAR sensors, reducing the range at which light may be received. Other conditions, such as

bright direct sunlight and/or high intensity headlights, may impede laser returns because the sensor's receiver rejects light outside the laser's operating range”)***, and***

- ***to provide an automation continuity command in response to the automation interruption event, wherein the automation continuity command includes instructions to delay, modify, or reschedule operations when the required sensor data identified by the data description is unavailable from the end point of the network*** (Levinson, paragraph 0153-0154, 0164-0167, wherein “Additional sensor recovery strategies may include accessing a map database to determine an alternative trajectory, route, and/or path that avoids one or more conditions that may be causing the anomalous sensor measurements. For example, where the sensor measurements of laser returns may be attributed to too much direct sunlight, alternative paths and/or trajectories may be determined as a sensor recovery strategy included in the store 3718. The planner 3722 may then select a new trajectory through the trajectory selection module 3724 that includes less direct sunlight based on building blocking the direct sunlight. As a result, the detection of the cause of the sensor anomaly may influence the selection of a sensor recovery strategy, thus modifying the operation of the AV system responsive to the sensor anomaly”).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo and Koda with Levinson to identify loss and/or malfunction of a sensor and provide a recovery or an alternative resolution because this would improve the reliability and safety of the system by modifying the vehicle operation when the required information for executing the plan is missing.

Claim 11 is rejected under the same rationale as claim 1.

4. ***Claim***(s) **2, 7, 12, and 17** is/are rejected under 35 U.S.C. 103 as being unpatentable over Oddo et al (Patent. No.: US 10,296,862 B1) in view of Koda et al (Pub. No.: US 2018/0113460) and Levinson et al (Pub. No.: US 2017/0123428) and Suiter et al (Patent. No.: US 9,821,910 B1).

As pre claim 2, claim 1 is incorporated and Oddo, Koda and Levinson do not further disclose **wherein the automation continuity circuit is further structured to provide the automation continuity command as a cancel command.** However, Suiter discloses **wherein the automation continuity circuit is further structured to provide the automation continuity command as a cancel command** (Suiter, col 13 lines: 29-34, wherein "For example, the scram order may include at least one of a "return to base" command, a cease "operation in a defined geospatial area" command, a "mission cancel" command, or a "cancel/ground" command similar to a kill switch").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo, Koda and Levinson with Suiter to provide a cancel command as claimed because this would improve the reliability and safety of the system by allowing to cancel the plan when continuing the execution is unsafe.

As pre claim 7, claim 1 is incorporated and Oddo, Koda and Levinson do not further disclose **wherein the automation interruption event comprises a trigger event interruption for a trigger event associated with the automated action plan.** However, Suiter discloses **wherein the automation interruption event comprises a trigger event interruption for a trigger event associated with the automated action plan** (Suiter, col 13 lines: 29-34, wherein "For example, the scram order may include at least one of a "return to base" command, a cease "operation in

a defined geospatial area" command, a "mission cancel" command, or a "cancel/ground" command similar to a kill switch").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo, Koda and Levinson with Suiter to provide a cancel command as claimed because this would improve the reliability and safety of the system by allowing to cancel the plan when continuing the execution is unsafe.

Claims 12 and 17 are rejected under the same rationale as claims 2 and 7.

5. ***Claim(s) 3 and 13*** is/are rejected under 35 U.S.C. 103 as being unpatentable over Oddo et al (Patent. No.: US 10,296,862 B1) in view of Koda et al (Pub. No.: US 2018/0113460) and Levinson et al (Pub. No.: US 2017/0123428) and Komatsu et al (WO 2019/130911 A1 wherein US 20210157324 A1 is used for mapping).

As pre claim 3, claim 1 is incorporated and Oddo, Koda and Levinson do not further disclose **wherein the automation continuity circuit is further structured to provide the automation continuity command as a restart command.** However, Komatsu discloses **wherein the automation continuity circuit is further structured to provide the automation continuity command as a restart command** (Komatsu, paragraph 0057, wherein "If it is determined that the unmanned vehicle 2 can resume normal traveling, the management device 4 transmits a restart command for traveling on the basis of the target traveling data to the stopped unmanned vehicle 2 via the communication system 7 (Step S7). Accordingly, the unmanned vehicle 2 travels on the basis of the target travel data").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo, Koda and Levinson with Komatsu to provide a cancel command as claimed because this would improve the reliability and safety of the system by allowing to resume performing the plan when the conditions are proper.

Claim 13 is rejected under the same rationale as claim 3.

6. ***Claim(s) 4-5 and 14-15*** is/are rejected under 35 U.S.C. 103 as being unpatentable over Oddo et al (Patent. No.: US 10,296,862 B1) in view of Koda et al (Pub. No.: US 2018/0113460) and Levinson et al (Pub. No.: US 2017/0123428) and Komatsu et al Yelland et al (Pub. No.: US 2013/0338856).

As pre claim 4, claim 1 is incorporated and Oddo, Koda and Levinson do not further disclose **wherein the automation continuity circuit is further structured to provide the automation continuity command as a stage command comprising a stage of the automated action plan at which the automated action plan is to be resumed.** However, Yelland discloses **wherein the automation continuity circuit is further structured to provide the automation continuity command as a stage command comprising a stage of the automated action plan at which the automated action plan is to be resumed** (Yelland, paragraph 0106, wherein "When the FCC is in the Loiter_State 518 and receives a Resume_Scenario_Command (which may be sent by the GVC), and the previous state was the Scenario_State 516, the FCC will transition its state to Scenario_State upon completion of the current lap of the loiter pattern. In this case, once changing to Scenario_State the vehicle will resume the scenario waypoints from where they were left when entering the Loiter_State").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo, Koda and Levinson with Yelland to provide a cancel command as claimed because this would improve the reliability and safety of the system by allowing to resume performing the plan when the conditions are proper.

As pre claim 5, claim 1 is incorporated and Oddo, Koda and Levinson do not further disclose **wherein the automation continuity circuit is further structured to provide the automation continuity command as a continuation command.** However, Yelland discloses **wherein the automation continuity circuit is further structured to provide the automation continuity command as a continuation command** (Yelland, paragraph 0106, wherein "When the FCC is in the Loiter_State 518 and receives a Resume_Scenario_Command (which may be sent by the GVC), and the previous state was the Scenario_State 516, the FCC will transition its state to Scenario_State upon completion of the current lap of the loiter pattern. In this case, once changing to Scenario_State the vehicle will resume the scenario waypoints from where they were left when entering the Loiter_State").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo, Koda and Levinson with Yelland to provide a cancel command as claimed because this would improve the reliability and safety of the system by allowing to resume performing the plan when the conditions are proper.

Claims 14-15 are rejected under the same rationale as claim 4-5.

7. ***Claim(s) 6 and 16*** is/are rejected under 35 U.S.C. 103 as being unpatentable over Oddo et al (Patent. No.: US 10,296,862 B1) in view of Koda et al (Pub. No.: US 2018/0113460) and Levinson et al

(Pub. No.: US 2017/0123428) and Komatsu et al Pack et al (Pub. No.: US 2006/0089765).

As pre claim 6, claim 1 is incorporated and Oddo, Koda and Levinson do not further disclose **wherein the automation interruption event comprises a vehicle shutdown event.** However, Pack discloses **wherein the automation interruption event comprises a vehicle shutdown event** (Pack, paragraph 0028, wherein "On detection of the disassociation (STEP 110), autonomous control is interrupted in favor of other operational modes (STEP 116). For example, on detection of the disassociation, a system according to an embodiment of the invention initiates a vehicle shutdown sequence (STEP 114). This can occur when, for example, the vehicle operator depresses a "panic button," thereby allowing a controlled, safe shutdown of the vehicle").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo, Koda and Levinson with Pack to handle shutdown events as claimed because this would improve the reliability and safety of the system by detecting events causing and/or resulting in shutdown of the vehicle.

Claim 16 is rejected under the same rationale as claim 6.

8. ***Claim(s) 8 and 18*** is/are rejected under 35 U.S.C. 103 as being unpatentable over Oddo et al (Patent. No.: US 10,296,862 B1) in view of Koda et al (Pub. No.: US 2018/0113460) and Levinson et al (Pub. No.: US 2017/0123428) and Komatsu et al Sant et al (Pub. No.: US 2019/0332105).

As pre claim 8, claim 1 is incorporated and Oddo, Koda and Levinson do not further disclose **wherein the automation interruption event comprises an operational interruption event comprising an indication of at least one of: a determination that a vehicle operating**

condition precludes completion of the automated action plan, a determination that a vehicle operating condition precludes continuation of the automated action plan, or a determination that a vehicle operating condition precludes execution of at least one aspect of the automated action plan. However, Sant discloses **wherein the automation interruption event comprises an operational interruption event comprising an indication of at least one of: a determination that a vehicle operating condition precludes completion of the automated action plan, a determination that a vehicle operating condition precludes continuation of the automated action plan, or a determination that a vehicle operating condition precludes execution of at least one aspect of the automated action plan** (Sant, paragraph 0055, wherein "In some embodiments, mission generator can include an independent monitoring system that checks each generated waypoint for potentials errors, checks the movement commands for infeasibility and checks the current telemetry of the movable object to ensure it is not attempting to execute catastrophic maneuvers. The monitoring system can alter the client device of any potential issues through the mobile-onboard library **310**B. If the issue escalates beyond a configurable limit, the monitoring system can shut down execution of the mission and report the failure back to the client device").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo, Koda and Levinson with Sant to handle shutdown events as claimed because this would improve the reliability and safety of the system by detecting events causing and/or resulting in unsafe operation of the vehicle.

Claim 18 is rejected under the same rationale as claim 8.

9. ***Claim(s) 9-10 and 19-20*** is/are rejected under 35 U.S.C. 103 as being unpatentable over Oddo et al (Patent. No.: US 10,296,862 B1) in view of Koda et al (Pub. No.: US 2018/0113460) and Levinson et al (Pub. No.: US 2017/0123428) and Zoox et al (WO 2017/079474 A2).

As pre claim 9, claim 1 is incorporated and Oddo, Koda and Levinson do not further disclose **wherein the automated action plan comprises a portion of a policy, wherein the policy comprises a continuity description, and wherein the automation continuity circuit is further structured to determine whether the automation interruption event has occurred in response to the continuity description.** However, Zoox discloses **wherein the automated action plan comprises a portion of a policy, wherein the policy comprises a continuity description, and wherein the automation continuity circuit is further structured to determine whether the automation interruption event has occurred in response to the continuity description** (Zoox, paragraph 0081-0082, wherein "Further, planner 1164 is configured to receive policy data 1130, perception engine data 1132, and localizer data 1134. Policy data 1130 may include criteria with which planner 1164 uses to determine a path that has a sufficient confidence level with which to generate trajectories, according to some examples. Examples of policy data 1130 include policies that specify that trajectory generation is bounded by stand-off distances to external objects (e.g., maintaining a safety buffer of 3 feet from a cyclist, as possible), or policies that require that traj ectories must not cross a center double yellow line, or policies that require trajectories to be limited to a single lane in a 4-lane roadway (e.g., based on past events, such as typically congregating at a lane closest to a bus stop), and any other similar criteria specified by policies. Perception engine data 1132 includes maps of locations of static objects and dynamic objects of interest, and localizer data 1134 includes at least a local pose or position").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo, Koda and Levinson with Zoox to achieve

the claimed limitation because this would improve the reliability and safety of the system by determining with a certain confidence when the plan will no longer proceed normally.

As pre claim 10, claim 1 is incorporated and Oddo, Koda and Levinson do not further disclose **wherein the automated action plan comprises a portion of a policy, wherein the policy comprises a continuity description, and wherein the automation continuity circuit is further structured to provide the automation continuity command in response to the continuity description.** However, Zoox discloses**wherein the automated action plan comprises a portion of a policy, wherein the policy comprises a continuity description, and wherein the automation continuity circuit is further structured to provide the automation continuity command in response to the continuity description** (Zoox, paragraph 0081-0082, wherein "Further, planner 1164 is configured to receive policy data 1130, perception engine data 1132, and localizer data 1134. Policy data 1130 may include criteria with which planner 1164 uses to determine a path that has a sufficient confidence level with which to generate trajectories, according to some examples. Examples of policy data 1130 include policies that specify that trajectory generation is bounded by stand-off distances to external objects (e.g., maintaining a safety buffer of 3 feet from a cyclist, as possible), or policies that require that traj ectories must not cross a center double yellow line, or policies that require trajectories to be limited to a single lane in a 4-lane roadway (e.g., based on past events, such as typically congregating at a lane closest to a bus stop), and any other similar criteria specified by policies. Perception engine data 1132 includes maps of locations of static objects and dynamic objects of interest, and localizer data 1134 includes at least a local pose or position").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing data of the invention to modify Oddo, Koda and Levinson with Zoox to achieve

the claimed limitation because this would improve the reliability and safety of the system by determining with a certain confidence when the plan will no longer proceed normally.

Claims 19-20 are rejected under the same rationale as claims 9-10.

Response to Arguments

Applicant's arguments filed on 08/24/2026 have been fully considered but they are now moot in light of the new grounds of rejection.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ario Etienne can be reached at ario.etienne@uspto.gov. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2457